eeco2
INSIGHT | ACTION | CONTROL

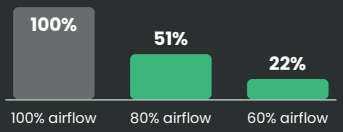
2-MINUTE SELF-CHECK

Is your cleanroom wasting energy?

Tick every statement that's true of your site. Each tick points to a saving you can usually make **while staying in grade**.

WHY AIRFLOW MATTERS

-20% airflow
≈ -49% fan power



Fan power ≈ airflow³ (fan affinity laws, ideal case). Small airflow cuts, big energy savings.

Operational efficiency

- [ ] Air change rates haven't been reviewed since commissioning.
- [ ] Fans run at full speed overnight and at weekends.

Sustainability & carbon

- [ ] We don't know what share of site energy our cleanrooms use.
- [ ] Our carbon targets leave GMP areas out of scope.

Digitalisation & data

- [ ] Particle data is logged for compliance, never used to control airflow.
- [ ] We can't see energy use per room or suite.

Compliance

- [ ] Energy projects stall because nobody's sure how QA will react.
- [ ] Energy measures aren't part of our Contamination Control Strategy.

Future-ready design

- [ ] Our next build or refit is designed around fixed, worst-case airflow.
- [ ] Our BMS can't adjust airflow to live room conditions.

COUNT YOUR TICKS

0–2
Well tuned
Prove it: benchmark your site against your peers.

3–5
Quick wins likely
See a real example overleaf.

6–10
Big opportunity
Worth a conversation. Scan the QR code overleaf.

Turn over: a real pharma example →

Guidance only, not a site assessment.

REAL EXAMPLE · PHARMACEUTICAL MANUFACTURING CAMPUS

A 28% air handling energy saving, with no added GMP risk

1 The challenge

Air change rates had been set conservatively to protect GMP compliance. Air handling used **over half the site's energy**, so it had to come down, without adding any GMP risk.

AIR HANDLING SHARE, 2025



of site energy



of site carbon

2 What we did

Modelled the site team's own operating data, recalculated air change rates to ISO 14644-16, and recommended six measures:

- **Demand-based setback** — Lower airflow when unoccupied
- **Air change reduction** — Set from particle data
- **Air pre-treatment** — Central fresh air conditioning
- **Heat recovery** — Heat from extract air
- **BMS optimisation** — Stable setpoint control
- **Efficient fans** — Higher-efficiency units

3 Projected results

PER YEAR, IDENTIFIED

28%	27%	29%	3.9 yrs
less air handling energy	less air handling carbon	lower air handling energy cost	simple payback

Status: Site Quality agreed the proposals in principle. Only measures adding no GMP risk were put forward.

How many did you tick? Let's talk them through.

Scan to talk through your ticks with an EECO2 engineer, and find out where your savings are.

SCAN TO TALK TO US

eeco2.com/contact

INSIGHT | ACTION | CONTROL

References: ISO 14644-16:2019 Cleanrooms and associated controlled environments – Energy efficiency; EU GMP Annex 1 (2022) Manufacture of Sterile Medicinal Products; fan affinity laws (ideal case). Guidance only, not a site assessment.